Prediction models for Progression-Free Survival in Breast Cancer Patients

Shivalika Chavan (sc5826), Kaveri Uberoy(kiu2103), Khue Nguyen (nnk2114)

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1. Introduction

Two sets of data about the progression-free survival of breast cancer patients were provided, originating from two similar studies. Dataset A comprises data from 2000 patients collected between 1990 and 1995. Dataset B was subsequently collected from 2000 to 2005, utilizing an identical study design, sample size, and set of variables.We built a prediction model for log-transformed PFS using the dataset A and evaluated the model's generalization of this model to dataset B.

2. Method

We cleaned both datasets and factorized the categorical variables including: menopause, race, tumor_grade, stage, surgery, chemo, hormonal_therapy, ER, PR and HER2. We explored the correlation of variables to determine multicollinearity by checking the correlation heatmap . We partitioned the dataset A into training (80%) and testing data (20%) and used repeated (n=5) 10-fold cross-validation to estimate out-of-sample prediction error. This approach repeatedly partitions the training data into training and validation sets, reducing variance compared to a single split. We evaluated seven candidate models: Linear Regression, K-Nearest Neighbors, Elastic Net (ENET), Principal Component Regression (PCR), Partial Least Squares (PLS), Generalized Additive Models (GAM), and Multivariate

Adaptive Regression Splines (MARS). Modeling was done using R version 4.5.1. Packages included "tidymodels" for cleaning data, "car" for assessing variance inflation factors (VIF), "caret" and "splines" for model training, "pls" for partial least squares model, "earth" for MARS model, "patchwork" for plot layout.

3. Results and discussion

3.1. Exploratory Data Analysis

Both dataset A and dataset B each contained data of 2000 patients with complete follow-up and no missing values. There are 15 predictors, including 5 continuous and ten categorical. The sample characteristics are described in Table 1.

Figure 1 and 2 show the correlation between variables in dataset A and B respectively . First, to look at data A, there are a few correlated variables, which intuitively make sense:

- `tumor_size`, `lymph_nodes`, and `stage` are highly positively correlated
- `age` and `menopause` are positively correlated
- `Ki67` (measures how fast cancer cells in a tumor are dividing) and `tumor_grade` (how abnormal cancer cells look) are positively correlated
- `Ki67` and `ER` (estrogen receptor) are negatively correlated
- `PR` (progesterone receptor) and `ER` are positively correlated
- `hormonal_therapy` and `ER` are positively correlated We checked variance inflation factors (VIF) values to assess the multicollinearity. All adjusted GVIF values were less than 2.5, indicating no multicollinearity. The highest value was observed for tumor_size (2.17), suggesting moderate but acceptable correlation with other variables (stage and lymph node), see Table 2. Thus, all variables were included in the modelling.

3.2. Model selection for dataset A

Based on the training data, most models performed very similarly, with mean RMSE values ranging from 0.7054 (PLS) to 0.7542 (GAM). (see Table 3). The similar performance across linear and nonlinear models suggests that the relationship between the predictors and log(PFS) is largely linear. PLS achieved the lowest mean error (0.7054), closely followed by PCR (Mean = 0.7066) and ENET (Mean = 0.709). ENET and PCR performing similarly to PLS suggests that dimensionality reduction provided little additional predictive power. ENET has the lowest maximum RMSE observed. Except for one outlier, the remaining RMSEs from cross-validation have a tight distribution, indicating this model is less variable and more stable. Because predictive accuracy was comparable across the top models, we prioritized interpretability. We selected the ENET model, which was trained using an explicit lambda grid to ensure consistent tuning across folds. The cross-validation process selected an alpha of 0, meaning the final model is a pure Ridge regression. While Ridge regression does not perform sparse feature selection (all variables are retained), it stabilizes the coefficients to handle multicollinearity, making it a highly robust and easily interpretable model for our clinical context. The coffecients for each variable of our selected model are shown in Table 4. The reported test error for this ENET model was 0.7445. (see Table 3)

3.3. Model testing and new model for dataset B

3.3.1. Model testing on dataset B

When applying the final Elastic Net model (trained on datA) to the new cohort (datB), the RMSE increased to 0.8422. Compared to the previous test RMSE of 0.7445 on the original data, this indicates a drop in predictive performance. This gap suggests that there may be underlying distribution shifts between the two patient populations (e.g., differences in demographics, clinical settings, or unmeasured variables), or that the model slightly overfitted the original cohort despite the use of Ridge regularization. A new model will be needed.

3.3.2. Train new model on dataset B

Since ENET had a tight distribution when using the original data, it seems the most promising (rather than training many different models). Additionally, it is then easier to compare what changed in the model.By comparing the coefficients of the old pre-2000 model to the new 2000-2005 model (see Table 5), several major predictive shifts become clear. The new model's intercept increased from 2.55 to 3.03, indicating that the baseline PFS increased over the decade. This makes sense since new and better interventions will have likely been more accessible. The impact of HER2 status (`HER21`) has flipped. In the older cohort, being HER2+ was a severe negative predictor for progression-free survival (-0.101). In the new cohort, this penalty essentially vanished (+0.009). Other predictors that flipped were all race categories, `menopause1`, `stage2`, and `tumor_grade2`. Other variables including ER and hormonal therapy stayed largely the same and continued to improve PFS, while stage 3, tumor grade 3, and tumor size decreased PFS. The gap in predictive power makes sense given these changes in cohort trends (see Table 5).

References

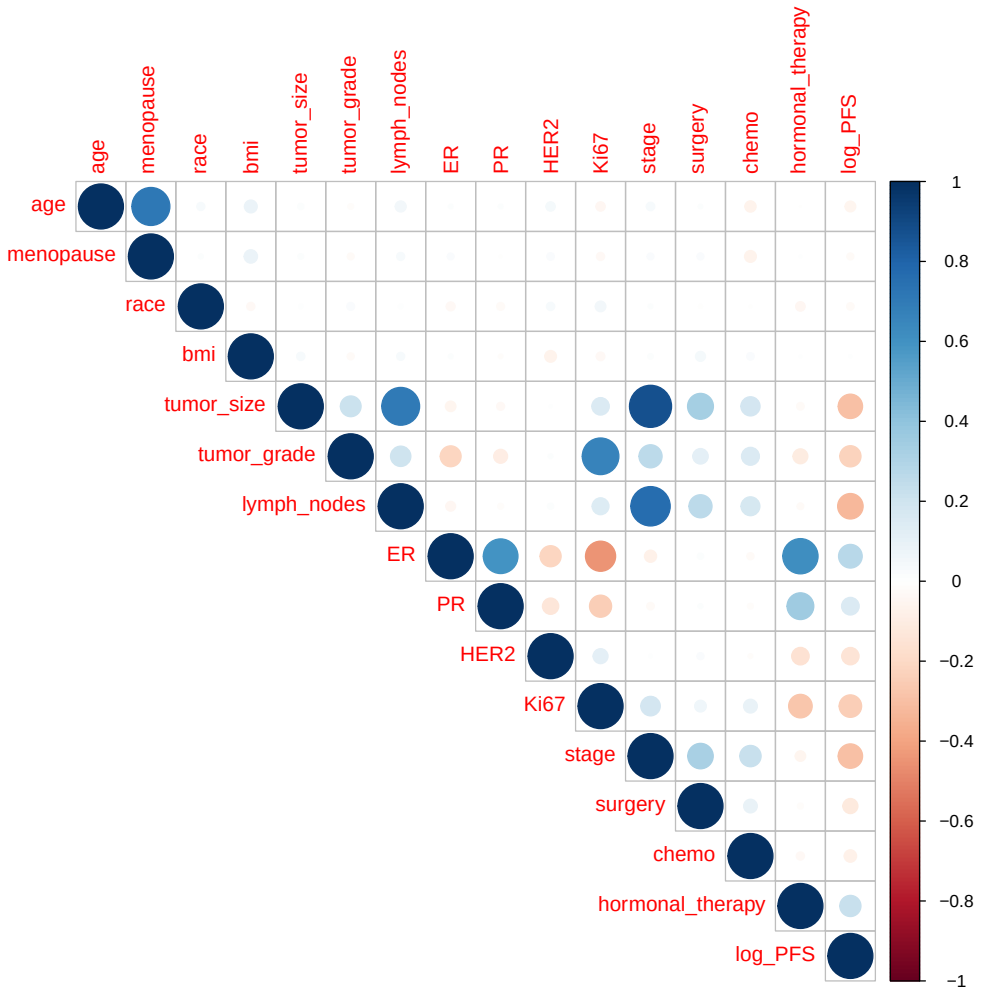

Figure 1: Correlation matrix of variables for dataset A

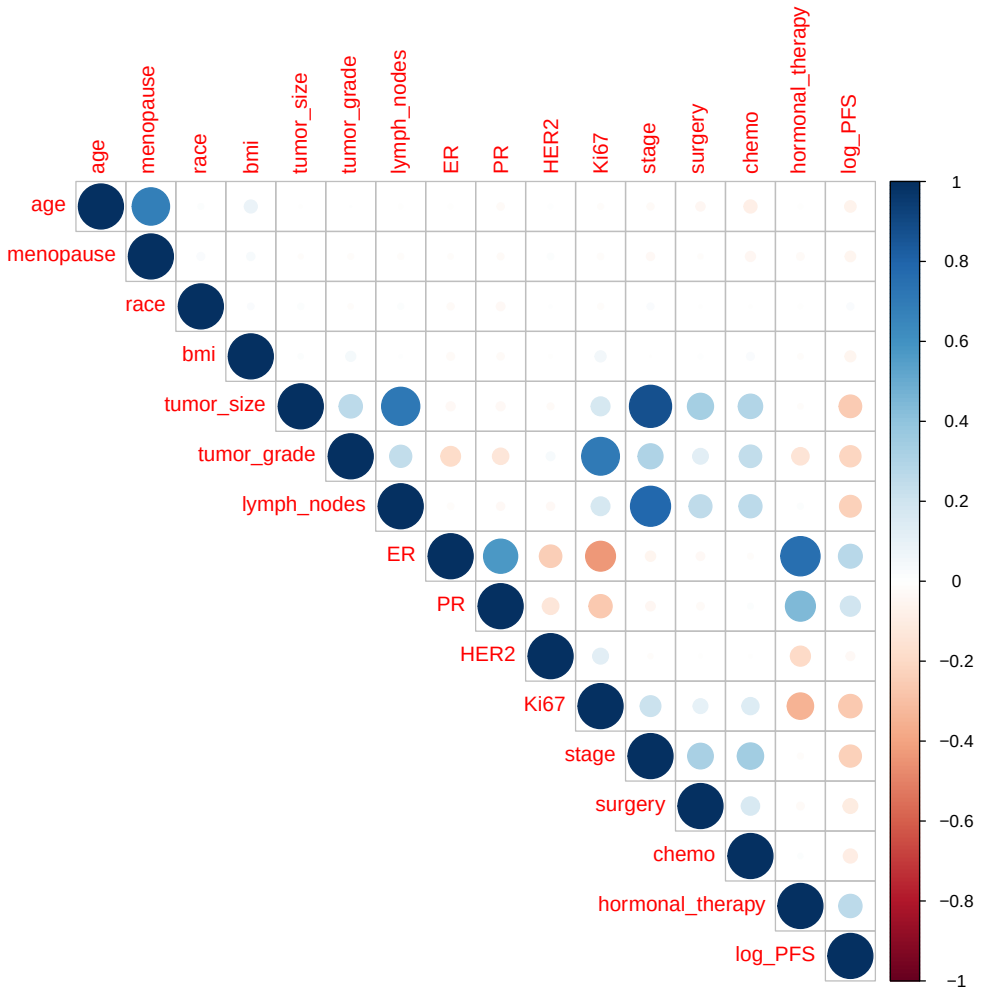

Figure 2: Correlation matrix of variables for dataset B

Table 1: Sample characteristics of Dataset A and Dataset B

Characteristic	**N = 2,000**	**N = 2,000**
age	55 (47, 62)	56 (49, 64)
menopause	NA	NA
0	689 (34%)	638 (32%)
1	1,311 (66%)	1,362 (68%)
race	NA	NA

Table 1: Sample characteristics of Dataset A and Dataset B *(continued)*

Characteristic	**N = 2,000**	**N = 2,000**
1	1,284 (64%)	1,097 (55%)
2	289 (14%)	369 (18%)
3	176 (8.8%)	220 (11%)
4	251 (13%)	314 (16%)
bmi	26.1 (23.0, 29.1)	27.3 (24.0, 30.9)
tumor_size	3.10 (2.10, 4.80)	2.80 (1.80, 4.20)
tumor_grade	NA	NA
1	383 (19%)	502 (25%)
2	871 (44%)	839 (42%)
3	746 (37%)	659 (33%)
lymph_nodes	2 (1, 4)	2 (0, 4)
ER	NA	NA
0	719 (36%)	652 (33%)
1	1,281 (64%)	1,348 (67%)
PR	NA	NA
0	1,039 (52%)	991 (50%)
1	961 (48%)	1,009 (50%)
HER2	NA	NA
0	1,556 (78%)	1,526 (76%)
1	444 (22%)	474 (24%)
Ki67	22 (14, 30)	20 (12, 29)
stage	NA	NA
1	504 (25%)	692 (35%)
2	893 (45%)	828 (41%)

Table 1: Sample characteristics of Dataset A and Dataset B *(continued)*

Characteristic	**N = 2,000**	**N = 2,000**
3	603 (30%)	480 (24%)
surgery	NA	NA
0	986 (49%)	1,117 (56%)
1	1,014 (51%)	883 (44%)
chemo	NA	NA
0	895 (45%)	702 (35%)
1	1,105 (55%)	1,298 (65%)
hormonal_therapy	NA	NA
0	1,066 (53%)	861 (43%)
1	934 (47%)	1,139 (57%)
log_PFS	2.19 (1.67, 2.71)	2.74 (2.25, 3.25)

Table 2: Variance Inflation Factors (VIF)

Variable	GVIF	Df	GVIF^(1/(2*Df))
age	2.120	1	1.456
menopause	2.115	1	1.454
race	1.031	3	1.005
bmi	1.020	1	1.010
tumor_size	4.712	1	2.171
tumor_grade	1.988	2	1.187
lymph_nodes	2.655	1	1.630
ER	2.538	1	1.593
PR	1.537	1	1.240
HER2	1.066	1	1.032
Ki67	2.192	1	1.481
stage	6.316	2	1.585
surgery	1.128	1	1.062
chemo	1.077	1	1.038
hormonal_therapy	1.635	1	1.279

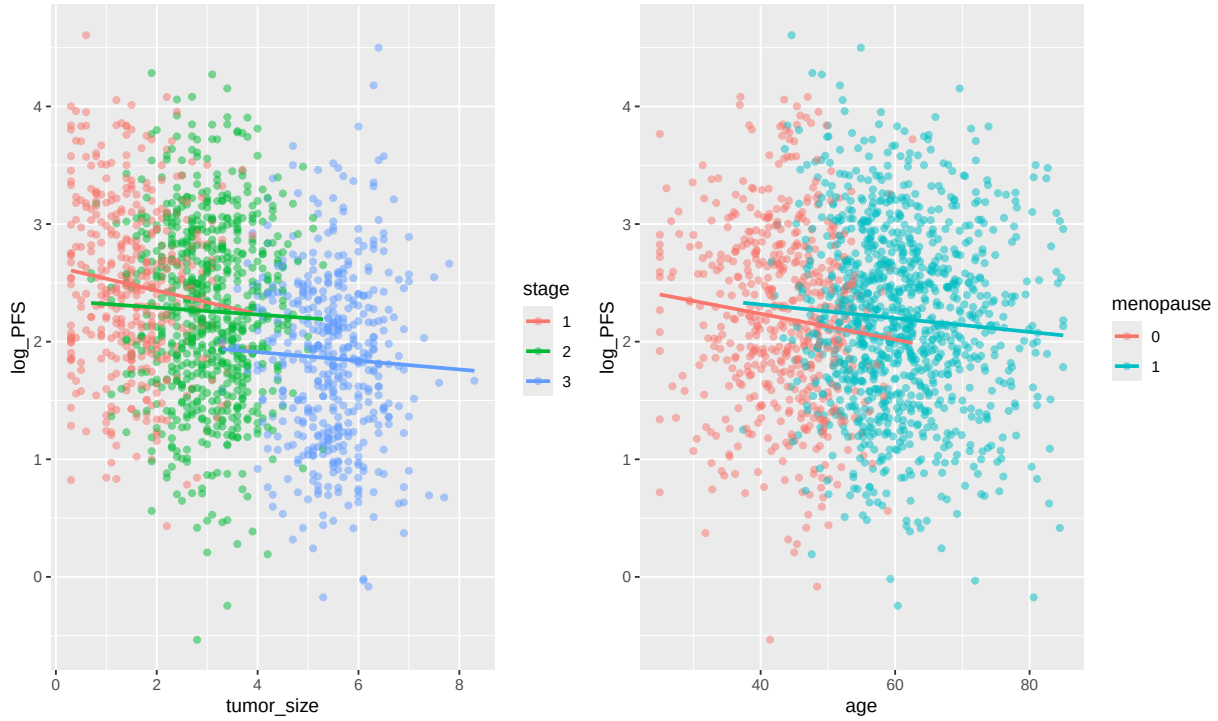

Figure 3: Interaction plot between tumor_size and stage, menopause and age

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
LM	0.7135	0.7310	0.7423	0.7402	0.7519	0.7604	0
KNN	0.7090	0.7168	0.7187	0.7200	0.7235	0.7332	0
ENET	0.6949	0.7083	0.7099	0.7090	0.7135	0.7165	0
PCR	0.6981	0.7010	0.7060	0.7066	0.7112	0.7208	0
PLS	0.6939	0.7000	0.7055	0.7054	0.7099	0.7206	0
GAM	0.7215	0.7457	0.7521	0.7542	0.7661	0.7909	0
MARS	0.7031	0.7214	0.7300	0.7305	0.7407	0.7590	0

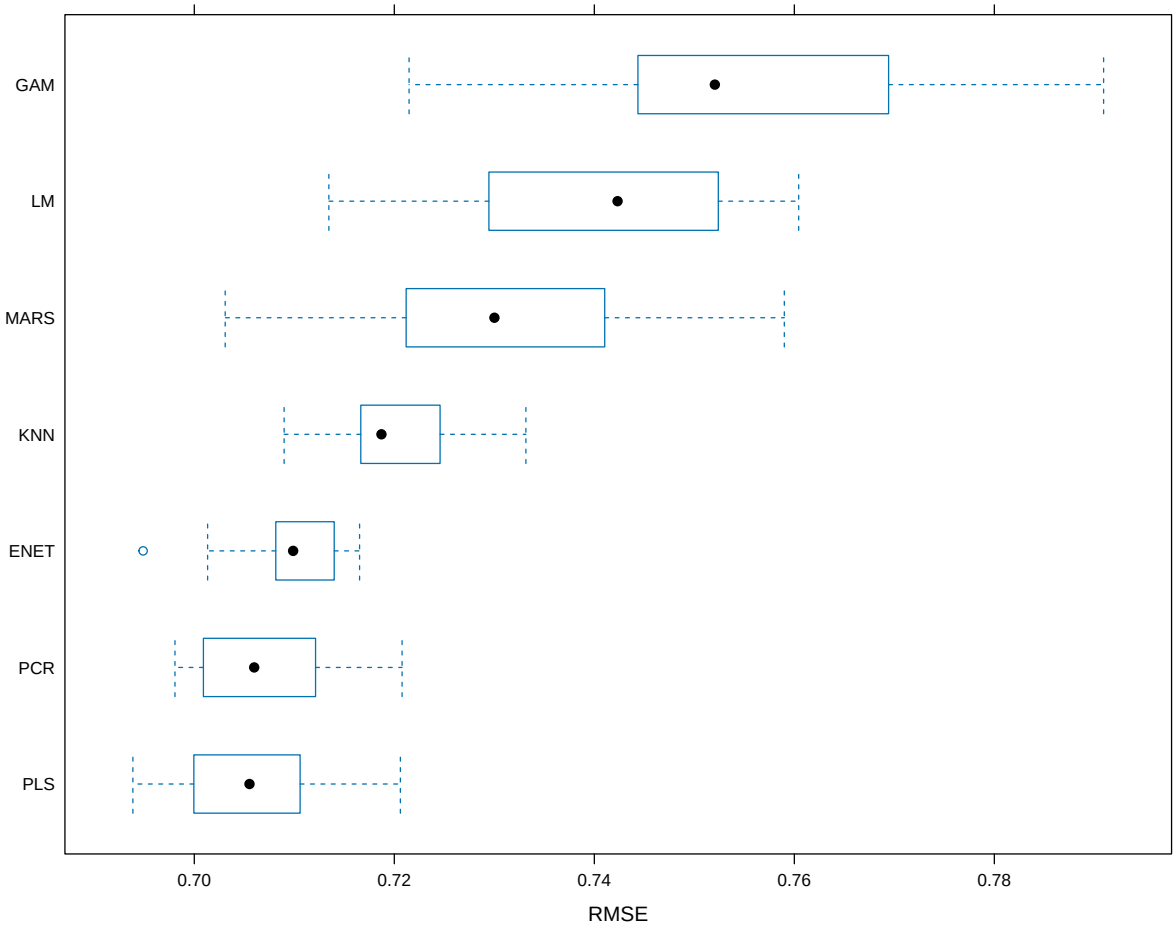

Figure 4: RMSE of tested models

Table 3: RMSE comparison for tested models

model	cv_rmse	test_rmse
GAM	0.730	0.741
ENET	0.709	0.744
PLS	0.705	0.745
KNN	0.720	0.746
PCR	0.707	0.748
LM	0.740	0.749
MARS	0.754	0.751

Table 4: Elastic Net Coefficients

predictor	coefficient
(Intercept)	2.550
ER1	0.129
hormonal_therapy1	0.106
stage3	-0.102
HER21	-0.101
tumor_grade3	-0.088
PR1	0.038
tumor_size	-0.034
surgery1	-0.033
race4	-0.032
lymph_nodes	-0.030
race3	0.028
race2	0.019
menopause1	0.015
chemo1	-0.012
tumor_grade2	-0.006
Ki67	-0.004
age	-0.002
bmi	0.000
stage2	0.000

Table 5: Coefficients comparision of two ENET models

predictor	coefficient_old	coefficient_new	difference
(Intercept)	2.5499	3.0341	-0.4842
ER1	0.1286	0.1117	0.0169
HER21	-0.1009	0.0086	-0.1096
Ki67	-0.0042	-0.0041	-0.0001
PR1	0.0384	0.0611	-0.0227
age	-0.0021	-0.0019	-0.0002
bmi	-0.0002	-0.0031	0.0028
chemo1	-0.0119	-0.0147	0.0027
hormonal_therapy1	0.1058	0.1224	-0.0166
lymph_nodes	-0.0300	-0.0158	-0.0142
menopause1	0.0149	-0.0220	0.0369
race2	0.0195	-0.0119	0.0314
race3	0.0284	-0.0284	0.0567
race4	-0.0318	0.0478	-0.0796
stage2	0.0000	-0.0105	0.0105
stage3	-0.1017	-0.0842	-0.0176
surgery1	-0.0325	-0.0476	0.0150
tumor_grade2	-0.0065	0.0568	-0.0633
tumor_grade3	-0.0881	-0.0822	-0.0059
tumor_size	-0.0343	-0.0293	-0.0050